

# Pushpendra Pandey

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## RESEARCH INTERESTS

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Dendroclimatology; Atmospheric circulation; Climate variability and change; Mountain meteorology; Hydro-climate extremes; Climate model evaluation.

## EDUCATION

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### Birbal Sahni Institute of Palaeosciences (BSIP)

Ph.D. in Climate Science

Apr 2022 – Present

Lucknow, India

- Thesis: *Climate Dynamics and Tree Growth Response Assessment in the Western Himalaya using Tree-rings.*

### University of Lucknow

M.A. in Geography (First Class)

2018 – 2020

Lucknow, India

- Master's thesis on downscaled climate data and extreme event analysis (Himalayan region).

### Pondicherry University

B.Tech. in Information Technology (First Class)

2008 – 2012

Puducherry, India

## APPOINTMENTS

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### Birbal Sahni Institute of Palaeosciences (BSIP)

Senior Research Fellow

Apr 2022 – Present

Lucknow, India

- Dendroclimate reconstructions (FMAM); circulation diagnostics; moisture attribution workflows; model evaluation.

## PUBLICATIONS

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### Journal articles

- [1] Pandey, P., Ashfaq, M., Lal, D., Shekhar, M., Singh, B. B., Ranhotra, P., Chand, P., & Bhattacharyya, A. (2026). Tree-ring evidence marks year 2022 as the driest spring season in nearly four centuries in the Western Himalayas. *Science of The Total Environment*, 1013, 181315. <https://doi.org/10.1016/j.scitotenv.2025.181315>
- [2] Jain, S., Chinthala, B. D., Pandey, P., et al. (2025). Forest-Stand Structure and Treeline Dynamics of Himalayan Fir over the Last Four Centuries in Kashmir, Western Himalaya. *Dendrochronologia*, 126442. <https://doi.org/10.1016/j.dendro.2025.126442>
- [3] Lal, D., Pandey, P., Shekhar, M., et al. Tree-Ring-Based Drought Reconstruction Reveals Increased Pre-Monsoon Droughts Over the Past Two Centuries in the Lug Valley, Kullu. *International Journal of Climatology*, e70104. <https://doi.org/10.1002/joc.70104>
- [4] Shekhar, M., Pandey, P., Singh, S., & Sharma, A. (2025). Monsoon dynamics and future projections in the Himalaya: Insights into sea surface temperature, sea level pressure relationships, and future flood risks. *Dynamics of Atmospheres and Oceans*, 110, 101558. <https://doi.org/10.1016/j.dynatmoce.2025.101558>
- [5] Shekhar, M., Sharma, A., Pandey, P., Sharma, A., & Dimri, A. P. (2024). Assessing the past and future dynamics of the Asian summer monsoon: Insights from palaeomonsoon synthesis and CMIP6 data. *Global Environmental Change Advances*, 2, 100004. <https://doi.org/10.1016/j.gecadv.2023.100004>

### Book chapter / edited volume

- [1] Chinthala, B. D., Singh, A., Shekhar, M., Tomar, N., Phulara, M., Yadav, A., Pandey, P., Ranhotra, P. S., Bhattacharyya, A., Joshi, R., Singh, C. P., Singh, S. P., Reshi, Z. A., & Joshi, R. (2023). Age-Girth Stand Structure of Himalayan Fir and Growth–NDVI Relationship in the Treeline Transects of Western Himalaya: An Ecological Perspective. In *Ecology of Himalayan Treeline Ecotone* (pp. 455–481). Springer Nature Singapore. [https://doi.org/10.1007/978-981-19-4476-5\\_19](https://doi.org/10.1007/978-981-19-4476-5_19)

### Under review

- [1] Nag, D., Pandey, P., Verma, S., et al. Early Holocene latitudinal variations in Monsoon–Westerly dynamics from the Northwest Himalaya. *Under review.*
- [2] Balraju, W., Shekhar, M., Pandey, P., Lal, D., Dhyani, R., & Sharma, A. Integrated dendrochemical analyses and standardized environmental indicators for reconstructing historical trace elemental pollution in Asia: a systematic review. *Under review.*

## METHODS & TOOLS

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- **Dendro** detrending; chronology building; EPS; nested PCA; climate-weighted PCs; site/species screening.
- **Atmos/Ocean** SWJ core latitude/strength; WD tracking & track density; EKE; 500–700 hPa GPH composites; ERA5; HadISST/OISST.
- **Lagrangian & recycling** FLEXPART; two-layer dynamic recycling; regime-conditioned source fractions.
- **Statistics** Bayesian linear models; GAM; block cross-validation; RE/CE; EVT (POT & block maxima); end-to-end uncertainty.
- **Computing** Python (xarray, dask, numpy/pandas, joblib, matplotlib/cartopy), R (dplR); HPC (OLCF Frontier, Andes).

## FIELDWORK

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- **Lahaul–Spiti (2024)** High-elevation transects 2700–3900 m; 116 cores (*Cedrus deodara*, *Pinus wallichiana*); north/east/southeast aspects.
- **Kashmir & Tugnath** Tree-ring sampling and chronology development across treeline and sub-alpine bands; moisture-sensitivity screening.

## WORKSHOPS & TRAINING

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- **ICTS-TIFR (May 2024)** Data Assimilation for Weather & Climate Models, Bengaluru.
- **NCAR (Jan 2024)** WRF Tutorial.
- **IUAC, New Delhi** HPC School.
- **GSI Training Institute (2023)** Remote Sensing & GIS, Hyderabad (Jan–Feb 2023).
- **RSAC Lucknow (2019)** Remote Sensing & GIS training.
- **University of Lucknow (2019)** Geospatial technology in water resources.

## EDITORIAL & REVIEW SERVICE

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- **Ad-hoc Reviewer** *Earth Systems and Environment* (Springer, 2026).

## SELECTED COLLABORATIONS

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- **Dr. Moetassim Ashfaq** — Oak Ridge National Laboratory, USA (HPC climate modelling; ERA5 moisture transport; WD dynamics).
- **Dr. Manmeet Singh** — Western Kentucky University, USA (diffusion-based precipitation super-resolution; S2S prediction).
- **Dr. Mukund Palat Rao** — Columbia University, USA (tree-ring field reconstructions; hydroclimate variability).
- **Dr. Kieran Hunt** — University of Reading, UK (Western Disturbance tracking; synoptic circulation diagnostics).
- **Prof. Zong-Liang Yang** — University of Texas at Austin, USA (land–atmosphere interactions; climate model evaluation).

## HONORS

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- **UGC-NET** All-India Rank 5 (99.99 percentile), June 2019.
- **UGC-NET** Qualified for Professorship, Dec 2018.

## COMMUNITY ENGAGEMENT

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- **AcSIR Science Club, BSIP** Vice President (2024–Present).
- **Young Earth System Scientists (YESS)** Member.

## PROFESSIONAL MEMBERSHIPS

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- **American Geophysical Union (AGU)** Student Membership.
- **American Meteorological Society (AMS)** Student Membership.
- **European Geosciences Union (EGU)** Student Membership.